1. Hypothesis Tests

1.1. Null & Alternative Hypothesis

- For a hypothesis test on the population mean μ, the **null hypothesis** $\mathbf{H*0}$ proposes a value $\mu*0$ for μ

  $H_0 : \mu = \mu_0$

- The **alternative hypothesis** $\mathbf{H*1}$ suggests the way in which μ might differ from $\mu*0$. H_1 can take three forms:

  $H_1 : \mu < \mu_0$, a one-tail test for a decrease

  $H_1 : \mu > \mu_0$, a one-tail test for an increase

  $H1 : \mu \neq \mu 0$, a two-tail test for a difference

- The **test statistic** is calculated from the sample. Its value is used to decide whether the null hypothesis should be rejected
- The **rejection** or **critical region** gives the values of the test statistic for which the null hypothesis is rejected
- The **acceptance region** gives the values of the test statistic for which the null hypothesis is accepted
- The **critical values** are the boundary values of the rejection region
- The **significance level** of a test gives the probability of the test statistic falling in the rejection region

To carry out a hypothesis test:

- Define the null and alternative hypotheses
- Decide on a significance level
- Determine the critical value(s)
- Calculate the test statistic
- Decide on the outcome of test depending on whether value of test statistic lies in rejection/acceptance region
- State the conclusion in words
- The test statistic z can be used to test a hypothesis about a population

$$z = \frac{\overline{x} - \mu}{\sqrt{\frac{\sigma^2}{n}}}$$

where μ is the population mean specified by H_0

- The critical values for some commonly used rejection regions:

Significance level	Two-tail	One-tail	
	$\mu = \mu_0$	$\mu > \mu_0$	$\mu < \mu_0$
10	± 1.645	1.282	-1.282
5	± 1.960	1.645	-1.645
2	± 2.326	2.054	-2.054
1	± 2.576	2.326	-2.326

1.2. Testing Different Distributions

- **Test for mean, known variance, normal distribution or large sample**

$X \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

- Use general procedure as outlined above
- **Test for mean, large sample, variance unknown**

  $X \sim N\left(\mu, \frac{s^2}{n}\right)$

  - Use the same procedure however must use unbiased estimate of the population variance, s

- **Test for large Poisson mean**

$X \sim N\left(\lambda, \frac{\lambda}{n}\right)$

- Use general procedure but must approximate normal distribution using the mean given
- Must apply continuity correction
- **Test for proportion, large sample** (Binomial distribution)

$$X \sim N\left(p, \frac{pq}{n}\right)$$

- Similar to Poisson approximation; using probability of success and applying continuity correction

1.3. Type I and Type II Errors

- A **Type I error** is made when a true null hypothesis is rejected
- A **Type II error** is made when a false null hypothesis is accepted
- **P(Type I error)** = significance level

	H_0 True	H_0 False
Reject H_0	Type I Error	Correct Rejection
Fail to Reject H_0	Correct Decision	Type II Error

- **Calculating P(Type II error):**
  - Firstly, calculate the acceptance region by leaving $\overline{x}$ as a variable and equating the test statistic to the significance level
  - Next, calculate the conditional probability that μ is now μ' and $\overline{x}$ is still in the acceptance region

P($\overline{x}$ is in acceptance region | $\mu = \mu'$)

Calculate this by substituting the limit of the acceptance region as $\overline{x}$ (calculated previously) and the new, given μ' into the test statistic equation and find the probability

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